

EDEC

Engineering Companion for Elite Dangerous

USER MANUAL

Operation · Privacy · Data flows · Unlock Guide

Version 21.164

This manual documents the implemented application. Network features are optional and enabled separately.

Developer: CMDR Forcer · GNU General Public License v3.0

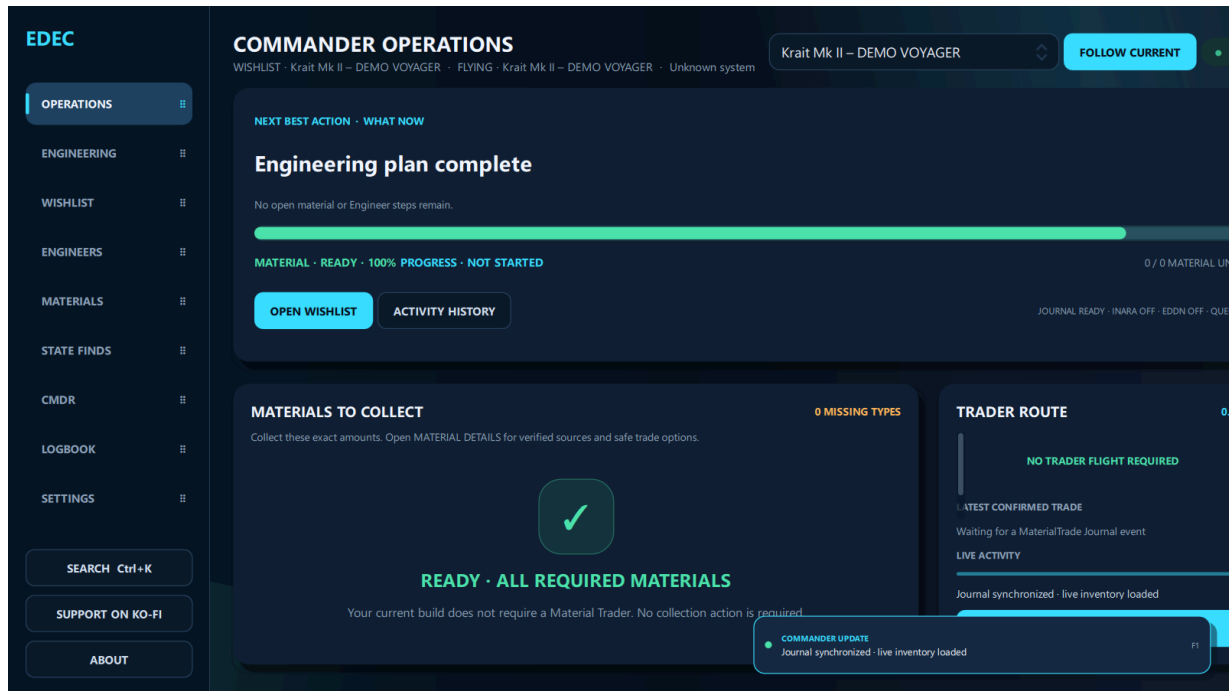
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1. Purpose and core principles

- EDEC is a local Engineering and Commander companion for Elite Dangerous. It combines Frontier Journal events, bundled reference catalogs and optional community services.
- Journal-first: Commander state and progress come from local Journal evidence and related snapshots.
- Offline-capable: engineering catalogs, material data, Engineer chains and reference coordinates are bundled.
- Opt-in networking: INARA, EDDN upload, EDDN Relay and Spansh actions are separately visible and controlled.
- Unknown history remains PENDING or UNKNOWN. Inventory deltas are applied only to a known full baseline.
- EDEC is an independent third-party tool and is not affiliated with Frontier Developments.

2. Quick start



Synthetic demonstration data; no real Commander account or credentials are shown.

- 1. Run START_APP.bat. Use INSTALL_REQUIREMENTS.bat once if Python packages are missing.
- 2. Verify the Journal directory under Settings. Journal Auto reads new evidence.
- 3. Select ship, module, blueprint, grade and optional experimental effect under Engineering.
- 4. Pin the plan to the Wishlist. EDEC calculates demand, inventory and protected surplus.
- 5. Follow Commander Operations for the next action, Engineer target, missing materials and trader guidance.
- 6. Enable INARA or EDDN separately under Settings > Connections. Nothing is sent without the relevant consent.

3. Navigation and main areas

- Operations prioritizes actions and material needs. Engineering builds plans. Wishlist combines multiple planned modules.
- Engineers contains the overview, Unlock Guide and Technology Broker recipes. Materials covers Raw, Manufactured and Encoded inventory.
- State Finds combines local and optional EDDN evidence. CMDR shows ranks, reputation, finances, current ship and squadron.
- Logbook stores a filtered local history and notes. Settings contains Journal, Connections, Diagnostics, tray and local-data controls.
- Navigation and CMDR cards can be reordered by drag-and-drop. Search, Ko-fi and About remain fixed.

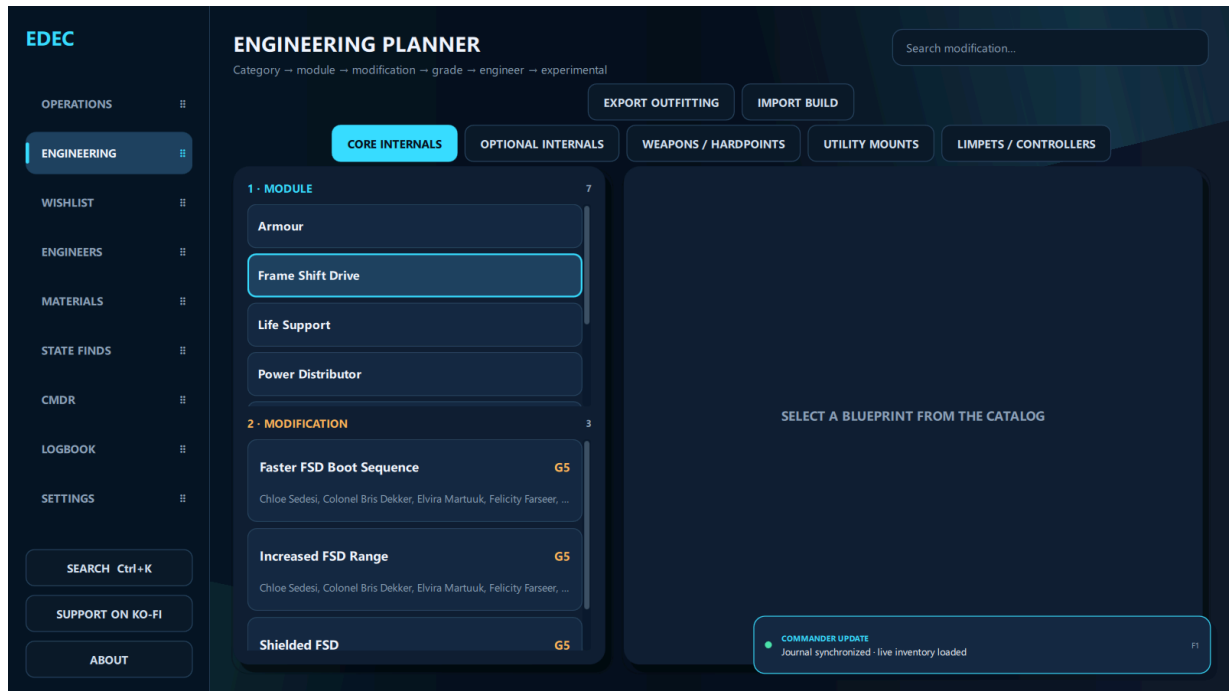
4. Data flow at a glance

- Local inputs: Elite Journal plus Cargo, Market, Outfitting, Shipyard, NavRoute, Module and Status snapshots.
- EDEC filters, correlates and persists local state. INARA receives approved personal Commander events only when enabled.
- EDDN receives schema-valid public galaxy and station data only when enabled. Its Relay can independently supply public State Finds evidence.
- Spansh is queried manually for station, trader and broker reference data; Commander inventory and Journal history are not uploaded to Spansh.

5. Local sources and storage

- Journal files are monitored by filename, modification time and size. Persistent cursors process new lines once and detect later growth in recently active files.
- Snapshot files establish authoritative baselines. A later full snapshot replaces intermediate delta calculations instead of adding them twice.
- Writable state is isolated under %LOCALAPPDATA%\EDEngineeringCompanion\profile-... and includes settings, plans, notes, cursors, receipts, queues and optional service configuration.
- API keys and Commander configuration stay local. Close EDEC before backing up or removing a profile directory.

6. Engineering, Wishlist and materials



Synthetic demonstration data; no real Commander account or credentials are shown.

- Engineering links a ship, module, blueprint, grade, experimental effect and recipe. Build import detects conflicts before replacement.
- Wishlist and fair-share planning combine requirements across ships and modules without allowing one plan to consume another plan's protected allocation.
- Material Details shows inventory, capacity, demand, estimated quantities, protected surplus, verified sources and safe trader options.
- A later full Materials snapshot replaces delta-derived interim state and prevents double counting.

7. Engineer Unlock Guide

The screenshot displays the EDEC interface with a sidebar on the left containing navigation links: OPERATIONS, ENGINEERING, WISHLIST, ENGINEERS (highlighted), MATERIALS, STATE FINDS, CMDR, LOGBOOK, and SETTINGS. Below these are buttons for SEARCH (Ctrl+K), SUPPORT ON KO-FI, and ABOUT.

The main content area is titled "ENGINEER NAVIGATION" and includes a search bar for "Engineer, system, module or blueprint..." and a filter set to "ALL". Below the search bar are three tabs: OVERVIEW, UNLOCK GUIDE (active), and TECH BROKERS.

Under the UNLOCK GUIDE tab, there are three summary boxes: ENGINEERS (25), UNLOCKED (3), and PENDING (1).

The selected engineer is Bill Turner, with a profile picture and a dropdown menu showing "Elvira Martuuk" and a "COPY SYSTEM" button. Below this is a progress bar labeled "PREREQUISITE -- STATUS -- NEXT STEP" with a "0 / 7" indicator.

The progress bar shows the following details:

- PREREQUISITE:** Selene Jean
- STATUS:** PENDING
- NEXT STEP:** Reach at least Grade 3 reputation with Selene Jean
- DESTINATION:** waiting for next navigable step
- CARGO:** 0 carried - 50 t still required - 64 t capacity

Below the progress bar is a "GUIDED UNLOCK CHAIN" section with a single step: "Unlock Selene Jean" (Reach at least Grade 3 reputation with Selene Jean), which is marked as "MISSING".

At the bottom, there is a "COMMANDER UPDATE" section showing "Journal synchronized - live inventory loaded" and a status "F1".

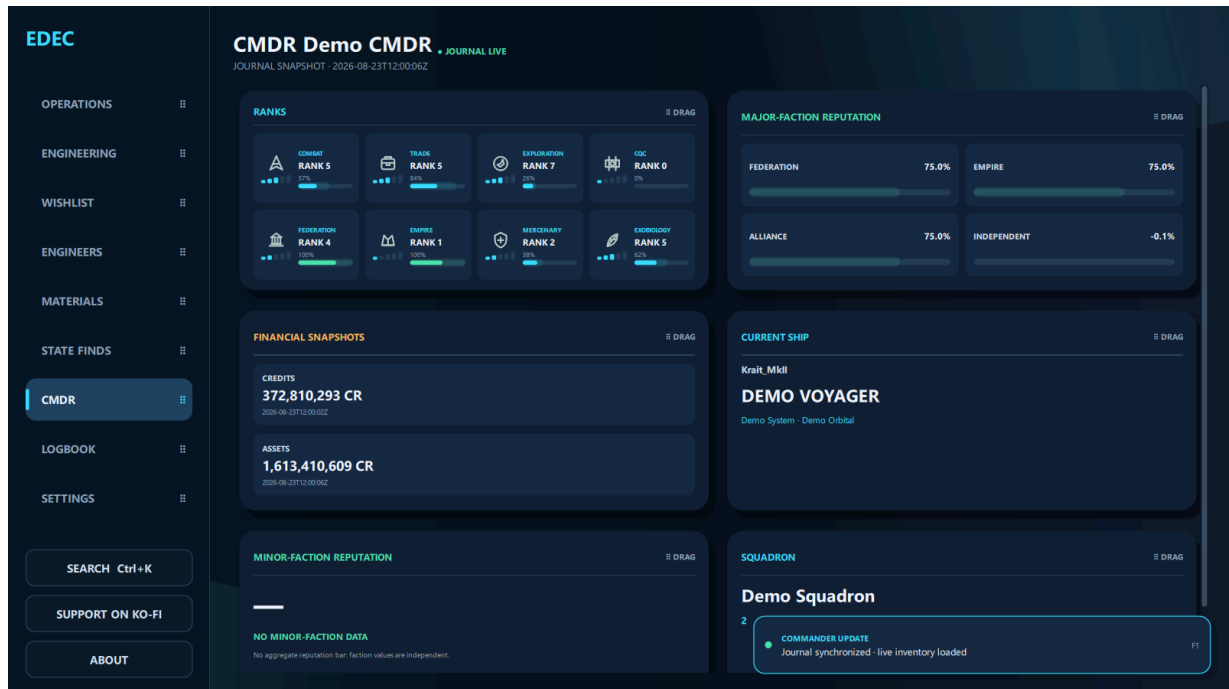
Synthetic demonstration data; no real Commander account or credentials are shown.

- Select an Engineer and activate Unlock Guide. Each chain shows prerequisite, evidence status and next verifiable action.
- READY means a supported step can be performed; PENDING means historical evidence is missing; LOCKED means prerequisites remain unmet.
- Evidence includes EngineerProgress, EngineerContribution, missions, docking/location, ranks, promotions and material/cargo state.
- The guide never marks a step complete solely because it appears plausible. Perform the next action and let the Journal confirm it.

8. Engineers, Technology Brokers, State Finds and Logbook

- Engineer status and rank come from Journal evidence; portraits and reference descriptions are local assets.
- Human and Guardian Technology Broker unlocks are tracked as one-time recipes against local inventory and Journal evidence.
- State Finds distinguishes LOCAL LIVE, LOCAL ENTERED, EDDN LIVE, RECENT REPORT and weaker candidate evidence. Time-sensitive signals expire.
- Logbook keeps a bounded local history and personal notes. Notes are never sent to INARA or EDDN.

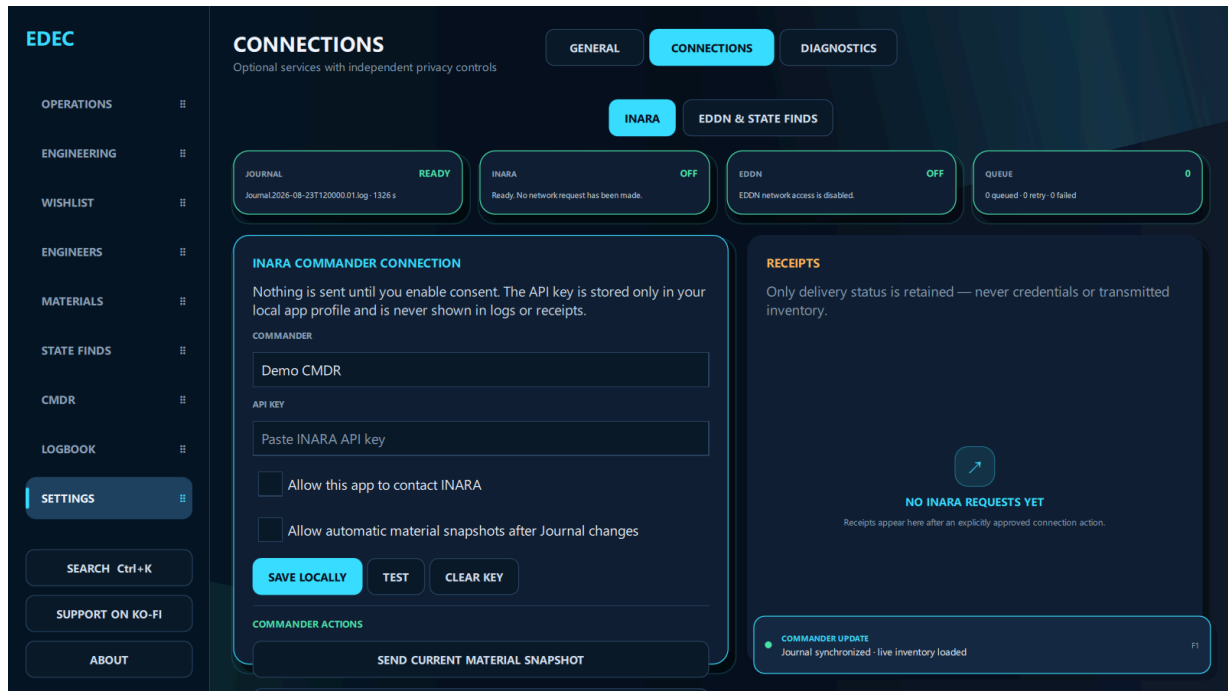
9. CMDR overview



Synthetic demonstration data; no real Commander account or credentials are shown.

- Compact cards show eight rank tracks, major-faction reputation, financial snapshots, current ship, minor-faction count and squadron.
- Credits come from consolidated Journal anchors; assets come from Statistics. No aggregate minor-faction reputation is invented.
- The CMDR page is a local view. Optional synchronization of related categories does not mean that the page is loaded from INARA.

10. Connections and consent



Synthetic demonstration data; no real Commander account or credentials are shown.

- INARA consent requires API key and Commander name; Auto Sync is a separate switch.
- EDDN consent, upload and Relay listener are separate controls. Receiving public Relay data does not require uploading.
- Spanish catalog updates are manual actions. Network status, receipts and failures remain visible.
- Disabling a service prevents new relevant requests; pending durable work follows the documented queue and retry rules.

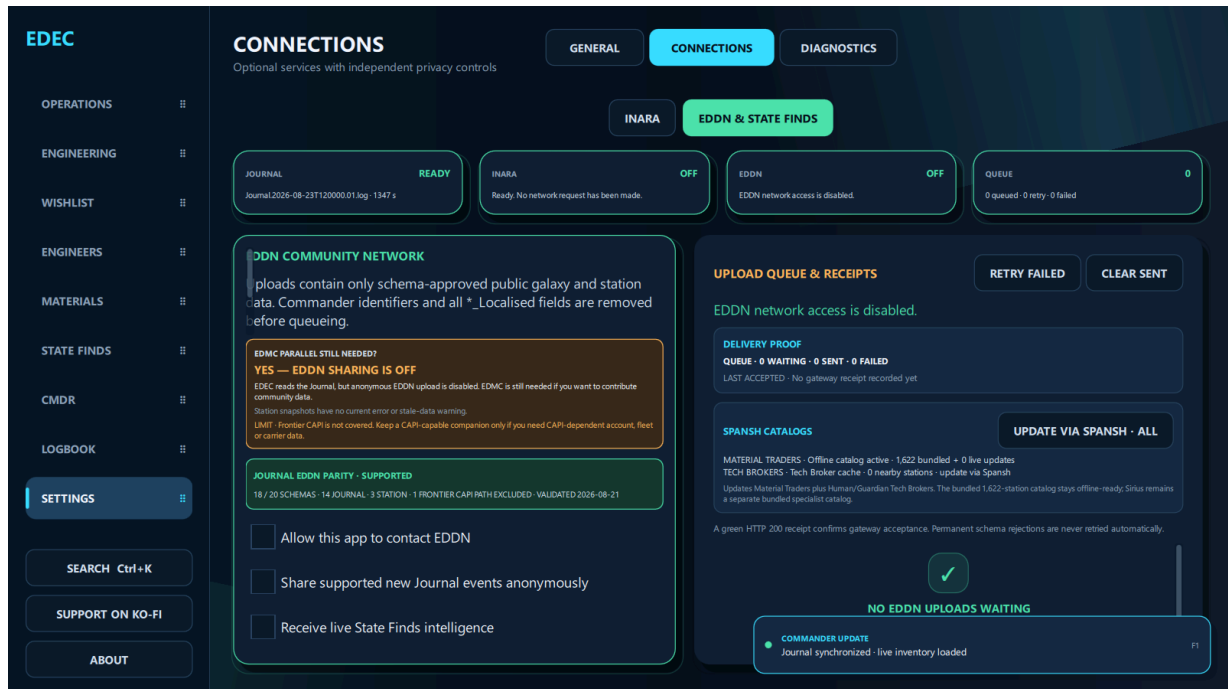
11. INARA - data sent

- Approved categories include inventory snapshots, travel, ranks, reputation, credits/statistics, fleet, ship and suit loadouts, missions, permits, combat and personal Community Goal progress.
- EDEC does not forward the complete Journal. Only whitelisted Live events with required fields enter a batch.
- Material, cargo, ShipLocker and module-storage deltas require a known full baseline. Full snapshots replace intermediate state.
- Beta, Alpha, Test and Legacy game sessions are rejected by the existing Live-data guard.

12. INARA - reads, timing and safeguards

- Read events: getCommanderProfile uses local caching; getCommunityGoalsRecent uses a long cache interval.
- Automatic collection window: 45 seconds. Minimum interval between automatic sends: 300 seconds, persisted across restart.
- Hard limit: at most 2 requests per minute. Batch size: 1-50 events. Successful SHA-256 fingerprints and receipts prevent duplicates.
- Retries use exponential backoff; rate-limit cooldown is long. The persistent pending queue has a visible bounded limit.

13. EDDN - public community data



Synthetic demonstration data; no real Commander account or credentials are shown.

- EDDN is a public real-time network for galaxy and station data. EDEC validates and compresses supported messages before the live gateway.
- The header contains a random local uploaderID, software/version and game version/build - not the Commander name.
- Private fields such as Commander, FID, PrivateGroup, Multicrew, personal reputation and squadron faction are removed. *_Localised fields are excluded.
- Gateway upload and ZeroMQ Relay listening are independent. Relay messages are validated and used only as transient public evidence.

14. EDDN - supported data

- Supported groups cover journal/1, settlement approach, codex, docking results, Fleet Carrier materials, FSS events, nav beacon/route, barycentre, organic scans, commodity/3, outfitting/3 with v2 fallback, and shipyard/2.
- scanorganic/1 sends schema-compliant Log/Sample observations; Analyse is excluded according to the schema rule.
- blackmarket/1 and fcmaterials_capi/1 remain intentionally unsupported because they require Frontier CAPI data unavailable to EDEC .

15. Spansh - data retrieved

- EDEC sends targeted station-search filters and reference coordinates to <https://www.spansh.co.uk/api/stations/search> only after a user action.
- Queries may include trader service/type, distance sorting and bounded page size. Results use system/station name, Market ID, coordinates, distance, pad/planetary properties, trader type and update time.
- Invalid or incomplete replies are rejected. Bundled catalogs remain available as an offline fallback.
- No Commander inventory, Journal series or automatic background telemetry is sent to Spansh.

16. External-service privacy matrix

- INARA: sends approved personal Commander state; reads profile and current Community Goals; identified by API configuration; opt-in.
- EDDN Gateway: sends public galaxy/station messages; receives receipts; uses random uploaderID; opt-in upload.
- EDDN Relay: sends nothing; receives public live messages; opt-in listener.
- Spansh: sends search filters/reference coordinates and reads station results after a manual action.
- GitHub and Ko-fi: EDEC sends no telemetry; the system browser opens only after an explicit click.

17. Offline operation, failures and queues

- DNS, connection and timeout failures are caught and shown while the application remains usable. Retryable failures retain their queue and use backoff.
- Invalid JSON or HTML responses produce explicit diagnostics instead of a false success. Schema-related terminal responses are not retried forever.
- Queues, cursors, receipts, rate-limit state and the INARA automatic-send interval persist across restart.
- A green receipt confirms gateway/API acceptance, not necessarily immediate rendering on an external website.

18. Frequently asked questions

- An unchanged INARA timestamp may be correct deduplication when no relevant inventory change occurred.
- Material or cargo deltas require a prior full snapshot baseline. Verify Journal Auto and INARA Auto after that baseline exists.
- No EDDN upload may mean the event has no supported public schema. Check consent, upload and schema status.
- PENDING unlock steps indicate missing historical evidence. Perform the next documented action and let the Journal update the state.
- UI scale supports a real 100%-150% range. With tray mode enabled, exit through the tray menu to stop the process.

Appendix A. Complete INARA upload whitelist

No.	Event	No.	Event
1	addCommanderCombatDeath	19	setCommanderInventory
2	addCommanderCombatInterdicted	20	setCommanderInventoryCargo
3	addCommanderCombatInterdiction	21	setCommanderInventoryMaterials
4	addCommanderCombatInterdictionEscape	22	setCommanderMissionAbandoned
5	addCommanderCombatKill	23	setCommanderMissionCompleted
6	addCommanderMission	24	setCommanderMissionFailed
7	addCommanderPermit	25	setCommanderRankEngineer
8	addCommanderShip	26	setCommanderRankPilot
9	addCommanderTravelCarrierJump	27	setCommanderRankPower
10	addCommanderTravelDock	28	setCommanderReputationMajorFaction
11	addCommanderTravelFSDJump	29	setCommanderReputationMinorFaction
12	addCommanderTravelLand	30	setCommanderShip
13	delCommanderShip	31	setCommanderShipLoadout
14	delCommanderSuitLoadout	32	setCommanderShipTransfer
15	resetCommanderInventory	33	setCommanderStorageModules
16	setCommanderCommunityGoalProgress	34	setCommanderSuitLoadout
17	setCommanderCredits	35	setCommanderTravelLocation
18	setCommanderGameStatistics	36	updateCommanderSuitLoadout

Additional cached read events: `getCommanderProfile` and `getCommunityGoalsRecent`.

Appendix B. Complete EDDN schema list

No.	Schema	No.	Schema
1	approachsettlement/1	10	fsssignaldiscovered/1
2	codexentry/1	11	journal/1
3	commodity/3	12	navbeaconscan/1
4	dockingdenied/1	13	navroute/1
5	dockinggranted/1	14	outfitting/2
6	fcmaterials_journal/1	15	outfitting/3
7	fssallbodiesfound/1	16	scanbarycentre/1
8	fssbodysignals/1	17	scanorganic/1
9	fssdiscoveryscan/1	18	shipyard/2

CAPI-only and intentionally not implemented: blackmarket/1 and fcmaterials_capi/1.

Appendix C. Sources, license and scope

Primary source	URL / purpose
EDEC source	https://github.com/CMDRForcer/ED-OPS - implemented version 21.164
INARA API	https://inara.cz/elite/inara-api-docs/ and /inara-api-devguide/
EDDN	https://github.com/EDCD/EDDN and its live schemas
Spansh	https://www.spansh.co.uk/ - station and galaxy reference service

EDEC is licensed under the GNU General Public License v3.0. This manual describes version 21.164 and may become outdated after later API, schema or UI changes.

Thanks to contributors, testers, bug reporters, translators and the Elite Dangerous community, and to EDCD/EDDN, INARA and Spansh for documentation, services and community resources.

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